STAT 325 - Design of Experiment
Homework 4

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3.4. A computer ANOVA output is shown below. Fill in the blanks. You may give bounds on the P-value.

One-way ANOVA

Source	DF	SS	MS	F	P
Factor	4	987.71	246.93	33.095	< 0.001
Error	25	186.53	7.4612		
Total	29	1174.24			

The missing values were computed as follows:

$$\begin{aligned}
\mathrm{DF}_{factor} &= 29 - 25 = 4 \\
\mathrm{SS}_{factor} &= 1174.24 - 186.53 = 987.71 \\
\mathrm{MS}_E &= \frac{186.53}{25} = 7.4612 \\
F_0 &= \frac{246.93}{7.4612} = 33.095 \\
P &= F_{\mathrm{cdf}}(\mathrm{Lower} = 33.09521,\ \mathrm{Upper} = 999,\ df_{\mathrm{num}} = 4,\ df_{\mathrm{den}} = 25) < 0.001
\end{aligned}$$

3.10. A product developer is investigating the tensile strength of a new synthetic fiber that will be used to make cloth for men's shirts. Strength is usually affected by the percentage of cotton used in the blend of materials for the fiber. The engineer conducts a completely randomized experiment with five levels of cotton content and replicates the experiment five times. The data are shown in the following table.

Cotton Weight Percent	**Observations**				
15	7	7	15	11	9
20	12	17	12	18	18
25	14	19	19	18	18
30	19	25	22	19	23
35	7	10	11	15	11

(a) Is there evidence to support the claim that cotton content affects the mean tensile strength? Use $\alpha = 0.05$.

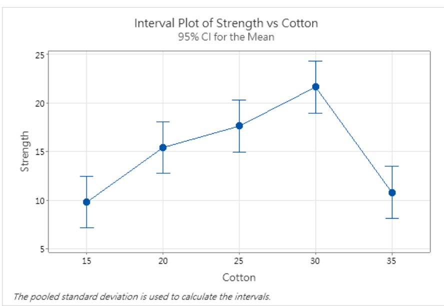
Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Cotton	4	475.8	118.940	14.76	0.000
Error	20	161.2	8.060		
Total	24	637.0			

Figure 1: Analysis of Variance

Using $\alpha = 0.05$ we see that the P-value is near 0 and we reject the null hypothesis where the null is that all the treatment means, or in this case the cotton weight percentage means, are the same. So we rejected that the means are the same and also we can visually see that if we plotted all the mean strengths from each cotton weight percentage group, they are clearly different from each other as well by a fair margin.

(b) Use the Fisher LSD method to make comparisons between the pairs of means. What conclusions can you draw?



(a) 95 % Confidence Interval of Means

Means

Cotton	N	Mean	StDev	95% CI
15	5	9.80	3.35	(7.15, 12.45)
20	5	15.40	3.13	(12.75, 18.05)
25	5	17.600	2.074	(14.952, 20.248)
30	5	21.60	2.61	(18.95, 24.25)
35	5	10.80	2.86	(8.15, 13.45)

Pooled StDev = 2.83901

(b) Confidence Intervals

The least significant difference for a balanced design is given as

$$\text{LSD} = t_{\alpha/2, N-a}\sqrt{\frac{2MS_E}{n}} \tag{1}$$

From computed values above we have the LSD as

$$\text{LSD} = \underbrace{t_{0.05/2, 25-5}}_{2.085963}\sqrt{\frac{2(8.060)}{5}} = 3.7455 \tag{2}$$

We then compute the $\binom{5}{2}$ pairs of difference for $|\overline{y}_{i.} - \overline{y}_{j.}| > \text{LSD}$ and conclude that the population mean μ_i, μ_j differ.

Fisher Individual Tests for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
20 - 15	5.60	1.80	(1.85, 9.35)	3.12	0.005
25 - 15	7.80	1.80	(4.05, 11.55)	4.34	0.000
30 - 15	11.80	1.80	(8.05, 15.55)	6.57	0.000
35 - 15	1.00	1.80	(-2.75, 4.75)	0.56	0.584
25 - 20	2.20	1.80	(-1.55, 5.95)	1.23	0.235
30 - 20	6.20	1.80	(2.45, 9.95)	3.45	0.003
35 - 20	-4.60	1.80	(-8.35, -0.85)	-2.56	0.019
30 - 25	4.00	1.80	(0.25, 7.75)	2.23	0.038
35 - 25	-6.80	1.80	(-10.55, -3.05)	-3.79	0.001
35 - 30	-10.80	1.80	(-14.55, -7.05)	-6.01	0.000

Simultaneous confidence level = 73.57%

Figure 2: $\overline{y}_{i.} - \overline{y}_{j.}$

From the *Difference of Means* column we can see that only the difference level of 35-15, 25-20 was not greater than the least significant value thus there is no mean difference between those two population mean pairs. However for the rest there is a significant difference between the group population means and this is also reflected in each 95 % confidence interval as 0 is not contained in the interval.

(c) Analyze the residuals from this experiment and comment on model adequacy.

From the *Normal Probability Plot* we can see that the residuals closely follows the red line which means that the residuals can be treated as normally distributed. Also from the *Residual vs Fitted Value plot* we can see that there no apparent pattern so we can assume constant variance across our data. Lastly, we can say that the data is collected is independent as the observation order of the residuals does not show consistent pattern.

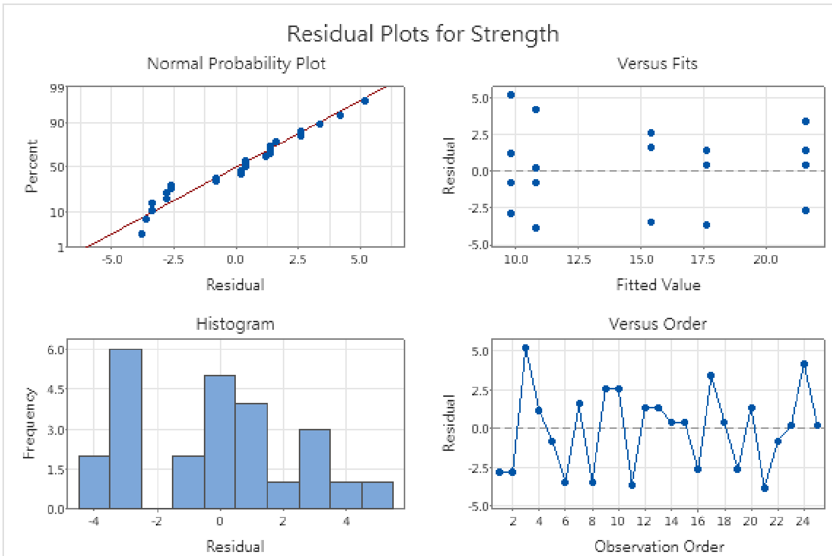


Figure 3: Residual Analysis on Strength

3.44. Suppose that four normal populations have means of $\mu_1 = 50$, $\mu_2 = 60$, $\mu_3 = 50$, and $\mu_4 = 60$. How many observations should be taken from each population so that the probability of rejecting the null hypothesis of equal population means is at least 0.90? Assume that $\alpha = 0.05$ and that a reasonable estimate of the error variance is $\sigma^2 = 25$.

Let the hypothesis test be

$$\begin{aligned} H_0 &: \mu_1 = \mu_2 = \mu_3 = \mu_4 \\ H_a &: \exists(j,i) : \mu_i \neq \mu_j \end{aligned} \tag{3}$$

Define the probability of rejecting the null hypothesis as the power

$$1 - \beta = P(\text{reject } H_0 \mid H_0 \text{ is False }) = P\{F_0 > F_{\alpha,a-1,N-a} \mid H_0 \text{ is false}\} \geq 0.90 \tag{4}$$

Rearranging for β yields

$$\beta \leq 0.1 \tag{5}$$

Then we can calculate the grand mean, treatment effects and treatment effect squared as

$$\bar{\mu} = \frac{1}{4}\sum_{1\leq i\leq 4} \mu_i = \frac{50+60+50+60}{4} = \frac{220}{4} = 55$$

$$\tau_i = \mu_i - \bar{\mu}$$

$$\tau_1 = 50 - 55 = -5, \quad \tau_2 = 60 - 55 = 5, \quad \tau_3 = 50 - 55 = -5, \quad \tau_4 = 60 - 55 = 5$$

$$\sum_{i=1}^{4} \tau_i^2 = (-5)^2 + (5)^2 + (-5)^2 + (5)^2 = 25 + 25 + 25 + 25 = 100$$

We can also compute the quantity Φ^2 for the non-centrality parameter δ

$$\Phi^2 = \frac{n \sum_{i=1}^{4} \tau_i^2}{a\sigma^2} = \frac{n \cdot 100}{4 \cdot 25} = \frac{100n}{100} = n \Rightarrow \Phi = \sqrt{n} \tag{6}$$

With a for loop in R we compute the f cumulative distribution for each n and degree of freedom until we pass the sufficient power level of 0.9.

n	$\Phi = \sqrt{n}$	$\nu_2 = 4(n-1)$	β	Power $= 1 - \beta$
3	1.732	8	0.37	0.63
4	2.000	12	0.18	0.82
5	2.236	16	0.07	0.93

We should take $n = 5$ observations from each population to achieve a power of at least 0.90. With 5 replicates per treatment, the power of the test is approximately 0.93, which satisfies the requirement.